PROGRAM 1:

```c
/*
 Francis Reuben R
 RVUN25CSE408
 B
 Program 1
 */
#include <stdio.h>
// Defining a macro named ADD(x,y) that computes the sum
#define ADD(x, y) ((x) + (y))
void main() {
    int num1, num2, sum; // User input: num1, num2. Output: sum
    printf("Enter first number: ");
    scanf("%d", &num1);
    // Prompt user for input
    printf("Enter second number: ");
    scanf("%d", &num2);
    // Call the ADD macro and store the result in sum
    sum = ADD(num1, num2);
    // Display the result
    printf("The sum of %d and %d is %d\n", num1, num2, sum);
}
```

OUTPUT:

```
metanaux@metanaux:~/Frank/PiC/f_Lab$ ./Prog1
Enter first number: 10
Enter second number: -9
The sum of 10 and -9 is 1
 80 metanaux@metanaux:~/Frank/PiC/f_Lab$ ./Prog1
Enter first number: 11
Enter second number: 55
The sum of 11 and 55 is 66
 80 metanaux@metanaux:~/Frank/PiC/f_Lab$ ./Prog1
Enter first number: -91
Enter second number: -9
The sum of -91 and -9 is -100
 80 metanaux@metanaux:~/Frank/PiC/f_Lab$
```

PROGRAM 2:

```c
/*
 Francis Reuben R
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 Program 1
 */
#include <stdio.h>
// Fn1: Swaps using a third variable (passbyreference)
void ThirdVar(int *a, int *b);
// Fn2: Swaps without a third variable (using arithmetic)
void Direct(int *a, int *b);
// Fn3: Swaps using pointers (implemented with bitwise XOR)
void Pointers(int *a, int *b);
void main() {
    int num1, num2;
    //  1. Test Fn1: With Third_variable
    printf(" 1. Swap WITH Third Variable (Fn1) \n");
    printf("Enter num1: ");
    scanf("%d", &num1);
    printf("Enter num2: ");
    scanf("%d", &num2);
    printf("Before swap: num1 = %d, num2 = %d\n", num1, num2);
    // Pass the addresses of the variables [cite: 393]
    ThirdVar(&num1, &num2);
    printf("After swap:  num1 = %d, num2 = %d\n", num1, num2);
    printf("\n\n\n");
    //  2. Test Fn2: Without a Third Variable
    printf(" 2. Swap WITHOUT Third Variable (Fn2) \n");
    printf("Enter num1: ");
    scanf("%d", &num1);
    printf("Enter num2: ");
    scanf("%d", &num2);
    printf("Before swap: num1 = %d, num2 = %d\n", num1, num2);
    Direct(&num1, &num2);
    printf("After swap:  num1 = %d, num2 = %d\n", num1, num2);
    printf("\n\n\n");
    //  3. Test Fn3: Pointers XORSwap
    printf(" 3. Swap using Pointers (XOR method) (Fn3) \n");
    printf("Enter num1: ");
    scanf("%d", &num1);
    printf("Enter num2: ");
    scanf("%d", &num2);
    printf("Before swap: num1 = %d, num2 = %d\n", num1, num2);
    Pointers(&num1, &num2);
    printf("After swap:  num1 = %d, num2 = %d\n", num1, num2);
}
// Fn1: With a Third variable
void ThirdVar(int *a, int *b) {
    int temp = *a; //Declare temp and assign num1 to temp
    *a = *b;       //Assign num2 to num1
    *b = temp;       // Assign temp to num2
}
//Fn2: Without a Third Variable
void Direct(int *a, int *b) {
    if (a == b) return;
    *a = *a + *b;
    *b = *a - *b; // *b = (*a + *b) - *b = *a (original)
    *a = *a - *b; // *a = (*a + *b)  *a = *b (original)
}
//Fn3: Pointers XORSwap
void Pointers(int *a, int *b) {
    if (a == b) return;
    *a = *a ^ *b;
    *b = *a ^ *b; // *b = (*a ^ *b) ^ *b = *a (original)
    *a = *a ^ *b; // *a = (*a ^ *b) ^ *a = *b (original)
}
```

OUTPUT:

```
metanaux@metanaux:~/Frank/PiC/f_Lab$ ./Prog2
 1. Swap WITH Third Variable (Fn1)
Enter num1: -9
Enter num2: 11
Before swap: num1 = -9, num2 = 11
After swap:  num1 = 11, num2 = -9



 2. Swap WITHOUT Third Variable (Fn2)
Enter num1: 15
Enter num2: 63
Before swap: num1 = 15, num2 = 63
After swap:  num1 = 63, num2 = 15



 3. Swap using Pointers (XOR method) (Fn3)
Enter num1: 12
Enter num2: 10
Before swap: num1 = 12, num2 = 10
After swap:  num1 = 10, num2 = 12
 80 metanaux@metanaux:~/Frank/PiC/f_Lab$ ./Prog2
 1. Swap WITH Third Variable (Fn1)
Enter num1: 2
Enter num2: 2
Before swap: num1 = 2, num2 = 2
After swap:  num1 = 2, num2 = 2



 2. Swap WITHOUT Third Variable (Fn2)
Enter num1: 12
Enter num2: 10
Before swap: num1 = 12, num2 = 10
After swap:  num1 = 10, num2 = 12



 3. Swap using Pointers (XOR method) (Fn3)
Enter num1: -9
Enter num2: -9
Before swap: num1 = -9, num2 = -9
After swap:  num1 = -9, num2 = -9
 80 metanaux@metanaux:~/Frank/PiC/f_Lab$
```

PROGRAM 3:

```c
/*
 Francis Reuben R
 RVUN25CSE408
 B
 Program 1
 */
#include <stdio.h>
// Fn1: Calculator logic using nested if-else
void nif(int choice, double num1, double num2);
// Fn2: Calculator logic using switch-case
void sw(int choice, double num1, double num2);
void main() {
    int choice;         // User input: choice
    double num1, num2; // User input: num1, num2
    // 1. Display the menu
    printf("Select an operation:\n");
    printf("1. Addition (+)\n");
    printf("2. Subtraction (-)\n");
    printf("3. Multiplication (*)\n");
    printf("4. Division (/)\n");
    printf("Enter your choice (1-4): ");
    // 2. Read the user's choice
    scanf("%d", &choice);
    printf("Enter two operands: ");
    scanf("%lf %lf", &num1, &num2);
    printf("\n");
    // 3. Execute the block of code corresponding to choice
    // - Call Fn1: Nested_If -
    printf("- Result (from Nested_If function) -\n");
    nif(choice, num1, num2);
    // - Call Fn2: Switch Case -
    printf("\n- Result (from Switch_Case function) -\n");
    sw(choice, num1, num2);
}
void nif(int choice, double num1, double num2) {
    if (choice == 1)
        printf("%.2lf + %.2lf = %.2lf\n", num1, num2, num1 + num2);
    else if (choice == 2)
        printf("%.2lf - %.2lf = %.2lf\n", num1, num2, num1 - num2);
    else if (choice == 3)
        printf("%.2lf * %.2lf = %.2lf\n", num1, num2, num1 * num2);
    else if (choice == 4) {
        // Handle division by ZERO
        if (num2 == 0)
            printf("Error! Division by zero is not allowed.\n");
        else
            printf("%.2lf / %.2lf = %.2lf\n", num1, num2, num1 / num2);}
    else
        printf("Error! Invalid choice. Please select from 1-4.\n");}
void sw(int choice, double num1, double num2) {
    switch (choice) {
        case 1: // Addition
            printf("%.2lf + %.2lf = %.2lf\n", num1, num2, num1 + num2);
            break;
        case 2: // Subtraction
            printf("%.2lf - %.2lf = %.2lf\n", num1, num2, num1 - num2);
            break;
        case 3: // Multiplication
            printf("%.2lf * %.2lf = %.2lf\n", num1, num2, num1 * num2);
            break;
        case 4: // Division
            // Handle division by ZERO
            if (num2 == 0)
                printf("Error! Division by zero is not allowed.\n");
            else
                printf("%.2lf / %.2lf = %.2lf\n", num1, num2, num1 / num2);
            break;
        default:
            printf("Error! Invalid choice. Please select from 1-4.\n");
    }}
```

OUTPUT:

```
metanaux@metanaux:~/Frank/PiC/f_Lab$ ./Prog3
Select an operation:
1. Addition (+)
2. Subtraction (-)
3. Multiplication (*)
4. Division (/)
Enter your choice (1-4): 1
Enter two operands: 10 -9

- Result (from Nested_If function) -
10.00 + -9.00 = 1.00

- Result (from Switch_Case function) -
10.00 + -9.00 = 1.00
 80 metanaux@metanaux:~/Frank/PiC/f_Lab$ ./Prog3
Select an operation:
1. Addition (+)
2. Subtraction (-)
3. Multiplication (*)
4. Division (/)
Enter your choice (1-4): 2
Enter two operands: 11 -6

- Result (from Nested_If function) -
11.00 - -6.00 = 17.00

- Result (from Switch_Case function) -
11.00 - -6.00 = 17.00
 80 metanaux@metanaux:~/Frank/PiC/f_Lab$ ./Prog3
Select an operation:
1. Addition (+)
2. Subtraction (-)
3. Multiplication (*)
4. Division (/)
Enter your choice (1-4): 3
Enter two operands: 99 -12

- Result (from Nested_If function) -
99.00 * -12.00 = -1188.00

- Result (from Switch_Case function) -
99.00 * -12.00 = -1188.00
 80 metanaux@metanaux:~/Frank/PiC/f_Lab$ ./Prog3
Select an operation:
1. Addition (+)
2. Subtraction (-)
3. Multiplication (*)
4. Division (/)
Enter your choice (1-4): 4
Enter two operands: 1444 78

- Result (from Nested_If function) -
1444.00 / 78.00 = 18.51

- Result (from Switch_Case function) -
1444.00 / 78.00 = 18.51
 80 metanaux@metanaux:~/Frank/PiC/f_Lab$ ./Prog3
Select an operation:
1. Addition (+)
2. Subtraction (-)
3. Multiplication (*)
4. Division (/)
Enter your choice (1-4): 9
Enter two operands: 12 12

- Result (from Nested_If function) -
Error! Invalid choice. Please select from 1-4.

- Result (from Switch_Case function) -
Error! Invalid choice. Please select from 1-4.
 80 metanaux@metanaux:~/Frank/PiC/f_Lab$
```